Kubernetes at Scale

Managing containerized microservices with automated rollouts, self-healing, and elastic cluster management.

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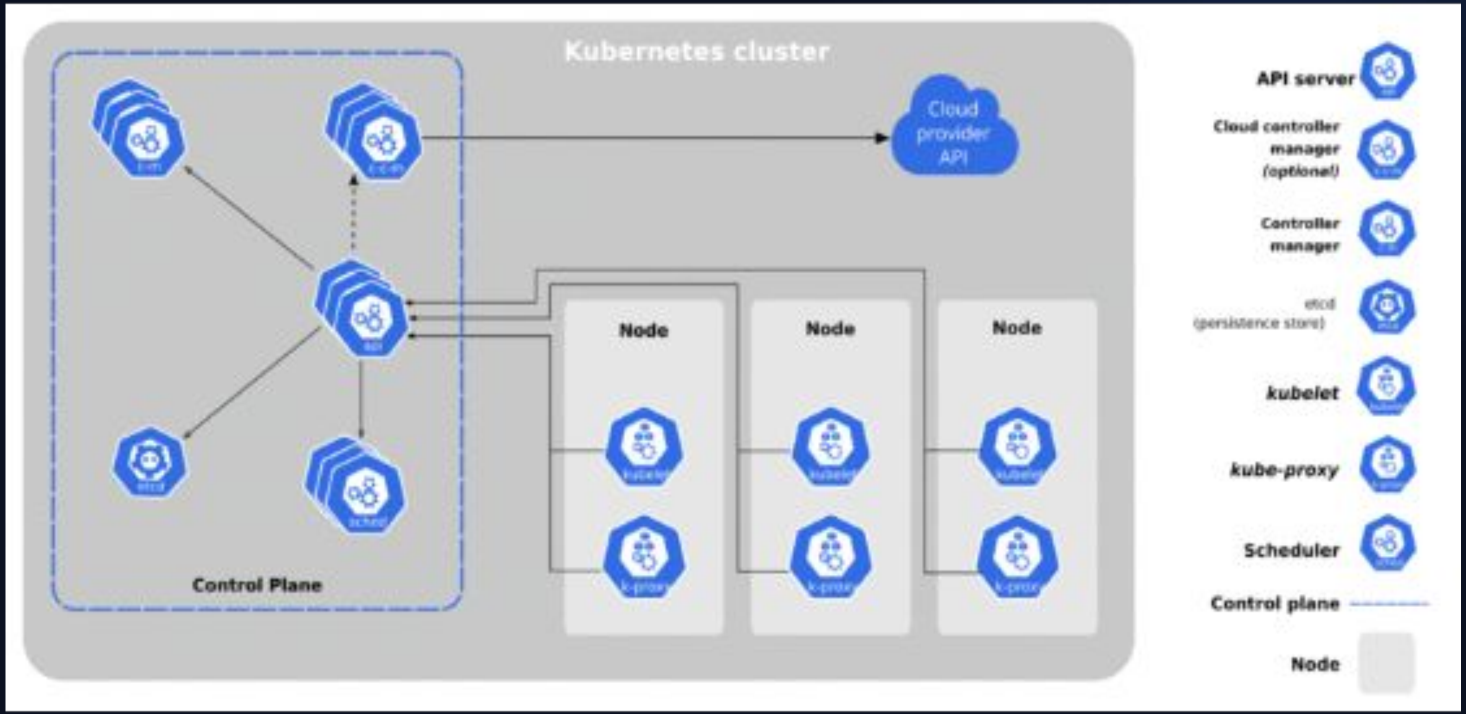
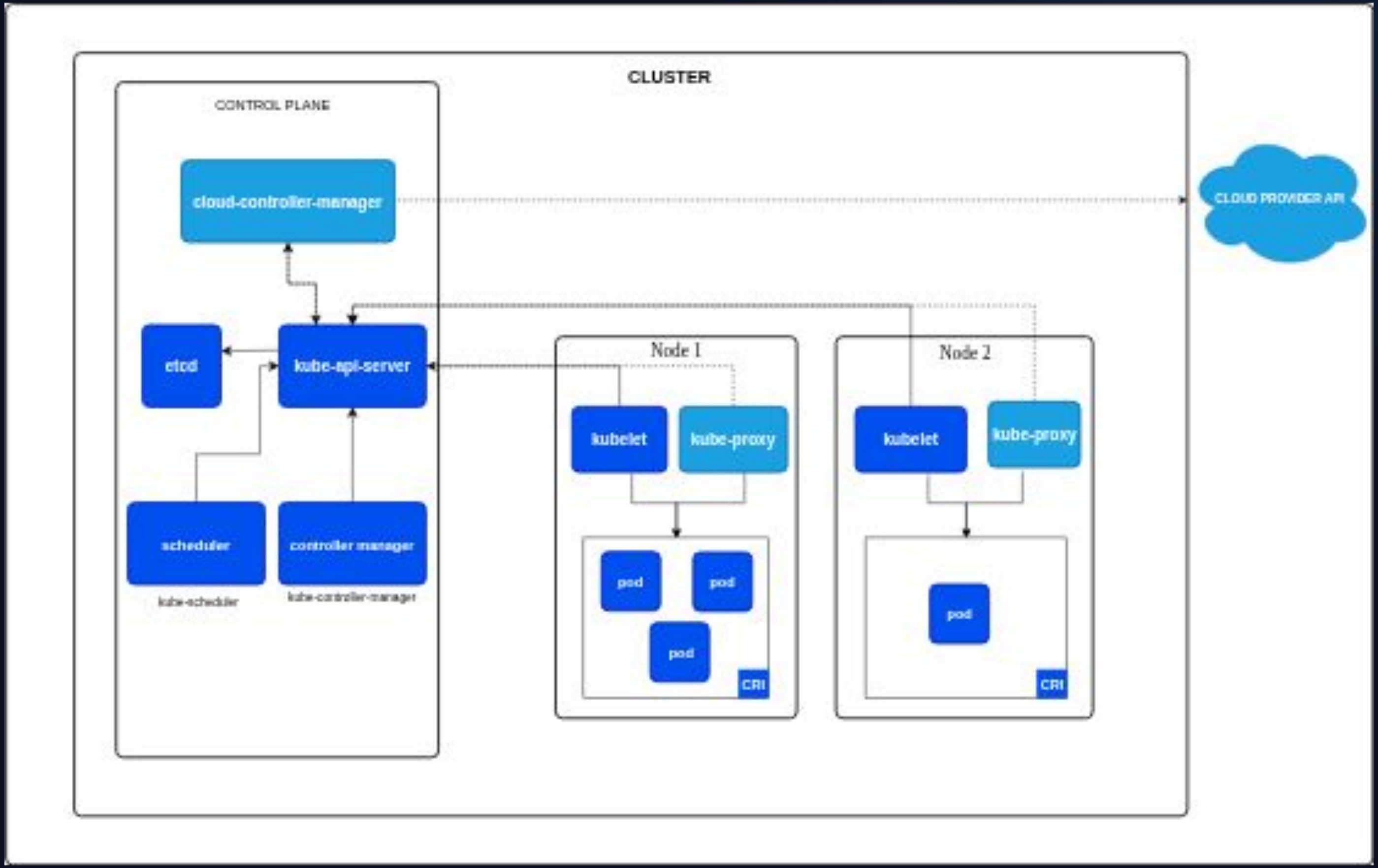
The Shift to Orchestration

- **Reconciliation Loop:** K8s constantly compares actual state against defined desired state.
- **Self-Healing:** Automatically restarts failed containers and reschedules pods when nodes die.
- **Horizontal Autoscaling:** Scale replica counts dynamically based on CPU, memory, or custom metrics.
- **Zero-Downtime Deploys:** Perform rolling updates and instant automated rollbacks.



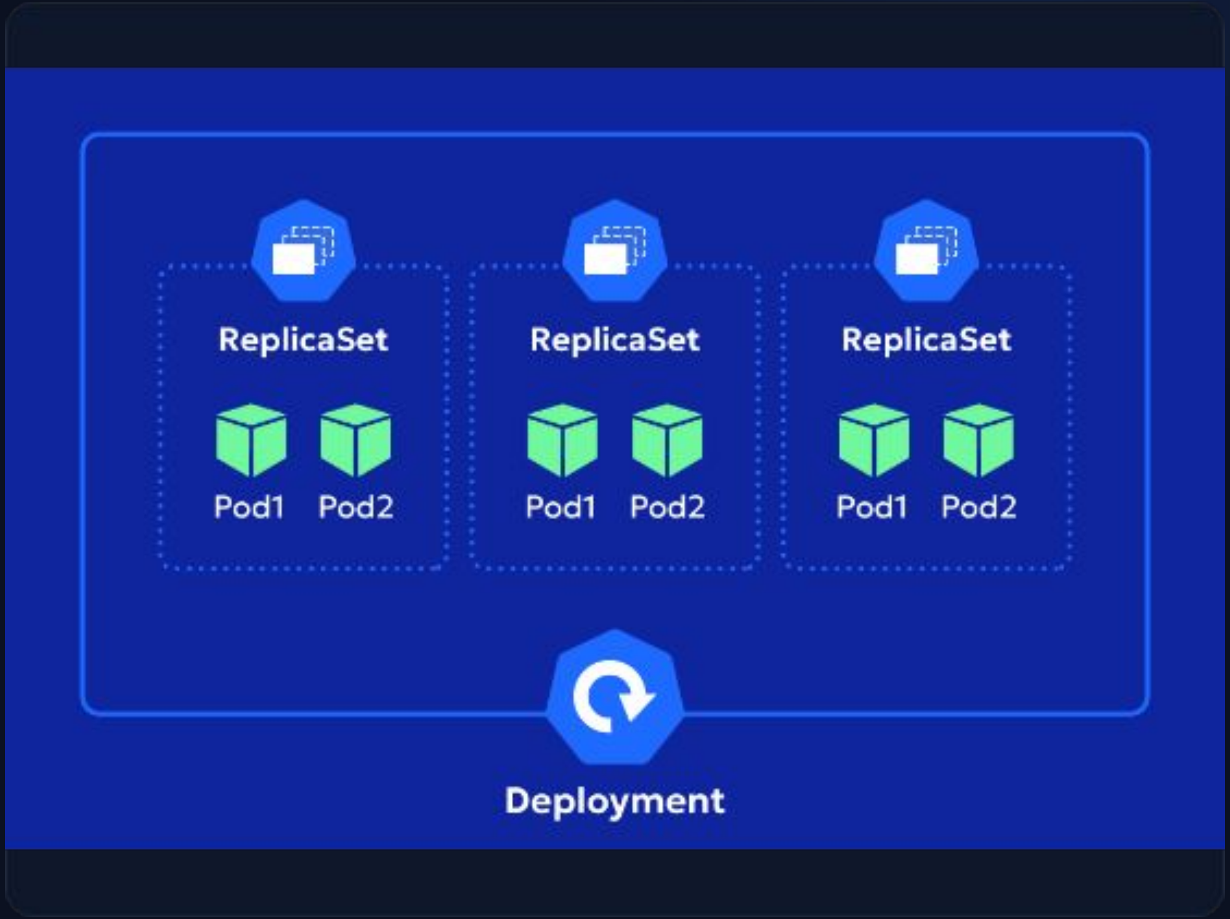
Kubernetes Architecture

- **Control Plane:** kube-apiserver, etcd (state storage), kube-scheduler, and controller-manager.
- **Worker Nodes:** Execute workloads via kubelet, kube-proxy, and container runtimes (e.g., containerd).
- **API Driven:** Standard REST endpoints controlled imperatively via kubectl or declaratively via YAML manifests.
- **Distributed State:** Strong consistency powered by the Raft consensus mechanism in etcd.



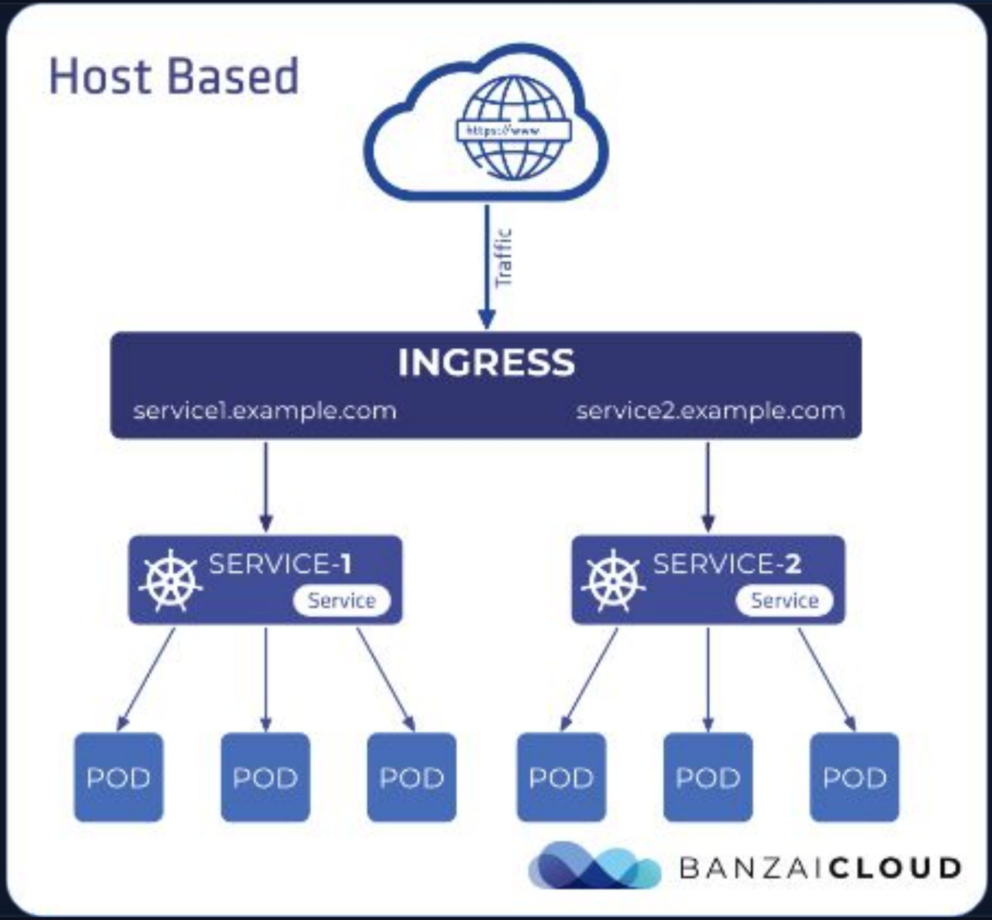
Pods & Workload Resources

- **The Pod Unit:** The smallest deployable object sharing IP address, IPC namespace, and storage volumes.
- **Deployments:** High-level abstraction managing ReplicaSets, declarative updates, and pod templates.
- **StatefulSets:** Provide stable, unique network identifiers and persistent disk bindings for stateful apps.
- **DaemonSets:** Guarantee that a single copy of a pod runs across all (or selected) cluster nodes.



Networking, Services & Ingress

- **Flat IP Model:** Every Pod receives a unique, routable cluster IP address without NAT required.
- **Services:** ClusterIP (internal routing), NodePort, and LoadBalancer abstractions for endpoint discovery.
- **Ingress Controller:** Layer 7 HTTP/HTTPS reverse proxy providing path-based routing and TLS termination.
- **Network Policies:** Granular firewall rules governing ingress and egress traffic between namespaces.



Storage, Config & Health

Config & Secrets

Decouple environment variables, files, and credentials from container images using ConfigMaps and encrypted Secrets.

Storage Abstraction

Dynamically provision cloud volumes using StorageClasses, PersistentVolumes (PV), and PersistentVolumeClaims (PVC).

Health Probes

Configure startupProbes, livenessProbes (restart trigger), and readinessProbes (traffic distribution filter).

Questions & Discussion

Thank you! You are now ready to present Docker & Kubernetes.

```
$ kubectl get pods --all-namespaces
```

Image Sources

https://enterprisedb.com/sites/default/files/images/kubernetes-diagram-2.gif

Source: www.enterprisedb.com

https://kubernetes.io/images/docs/kubernetes-cluster-architecture.svg

Source: kubernetes.io

https://zesty.co/wp-content/uploads/2025/02/Replica-Set.png

Source: zesty.co

https://outshift-headless-cms-s3.s3.us-east-2.amazonaws.com/blog/k8s-ingress/ingress-host-based-1.png

Source: outshift.cisco.com